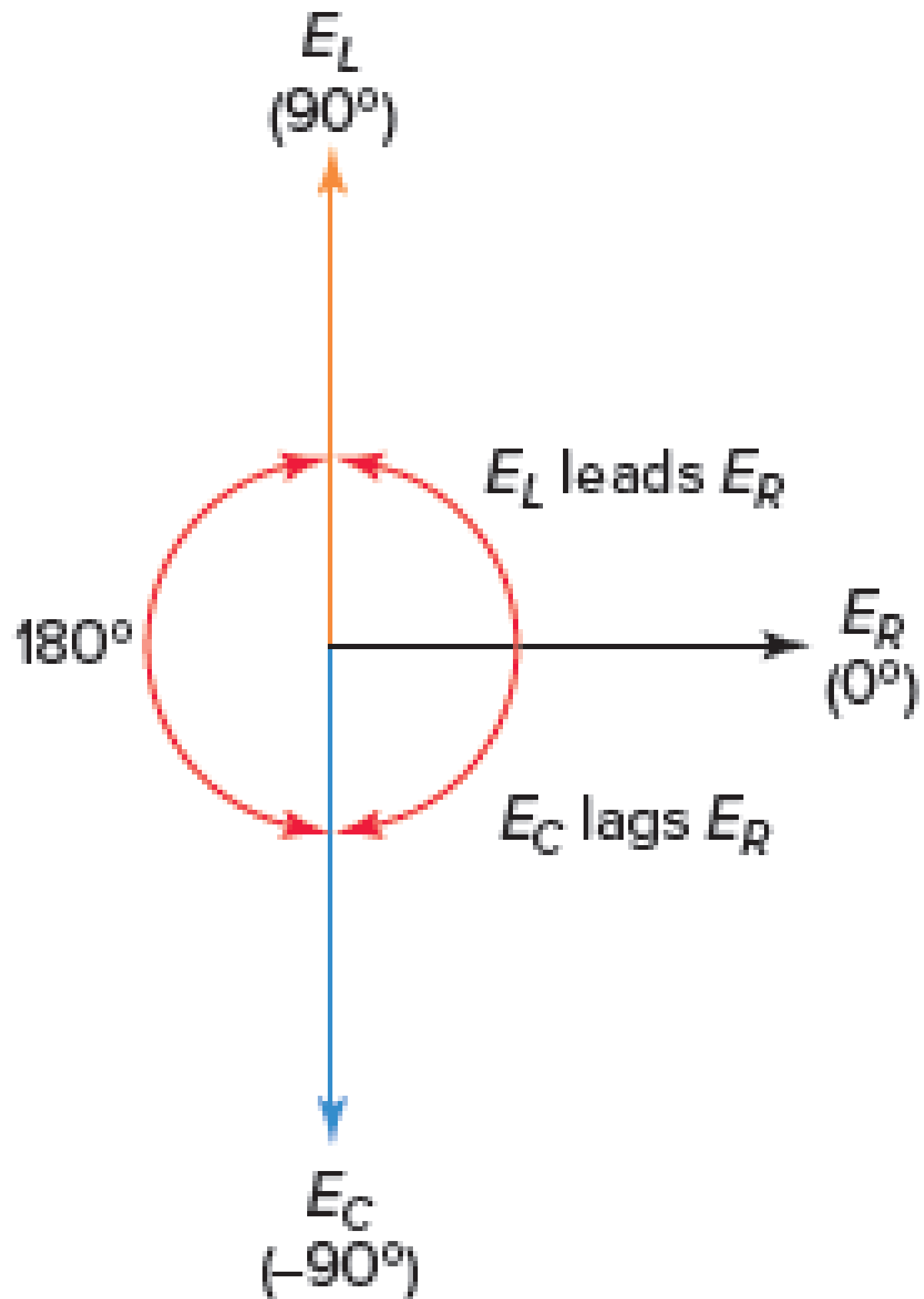
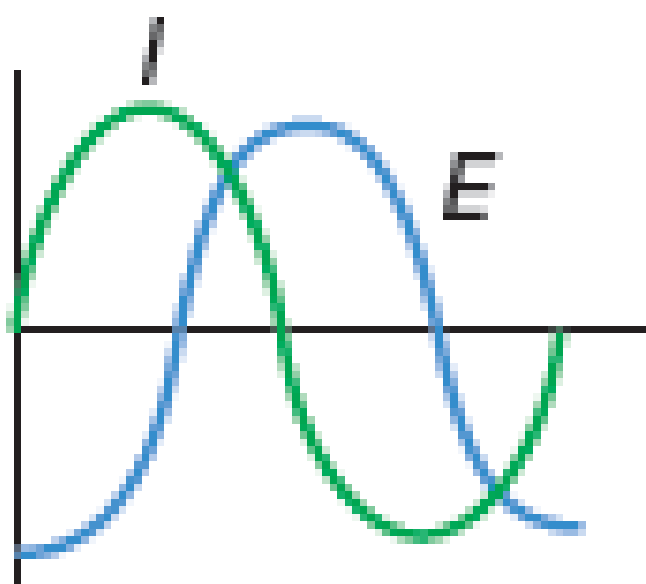
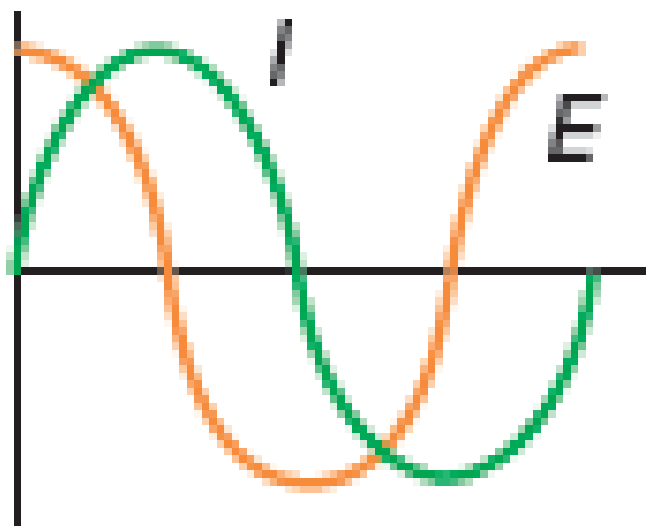
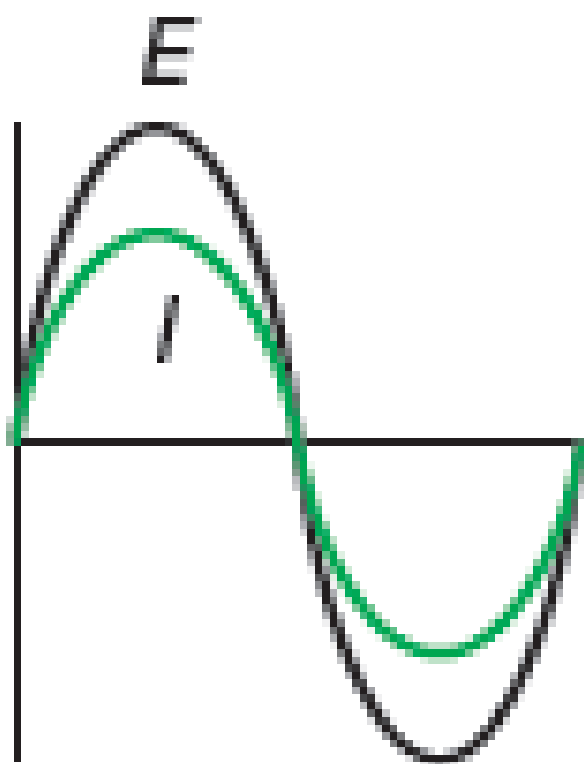
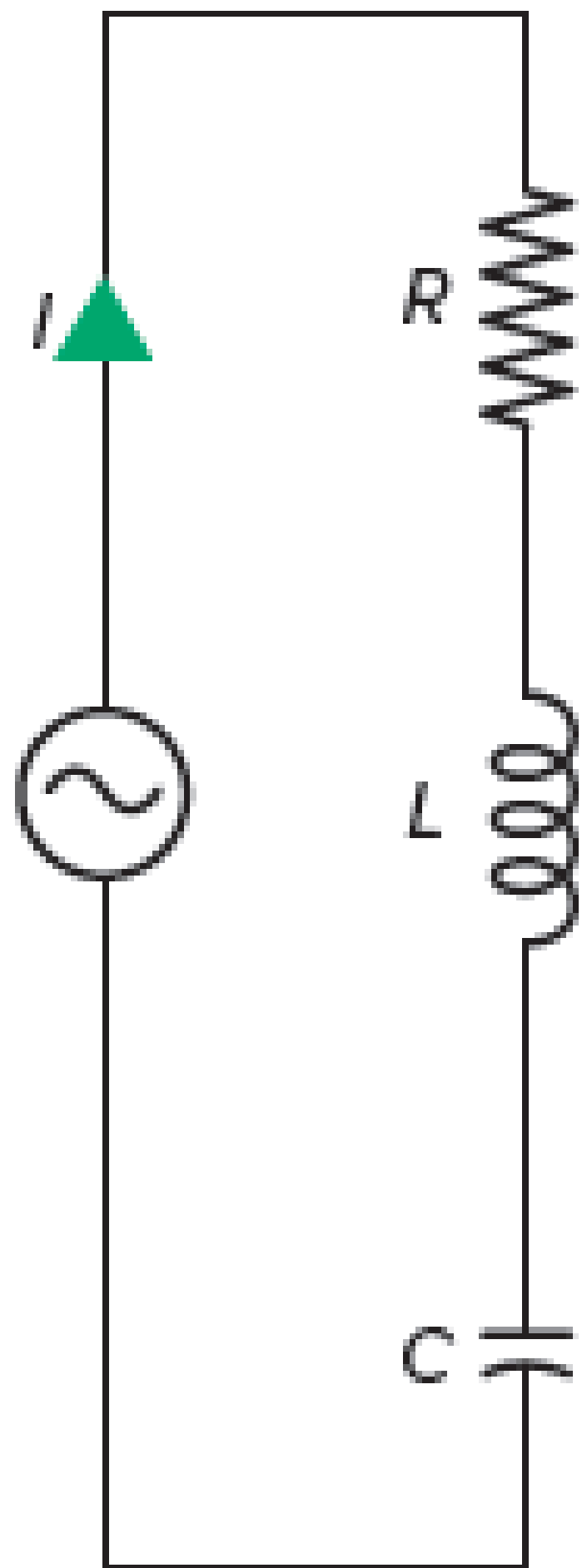
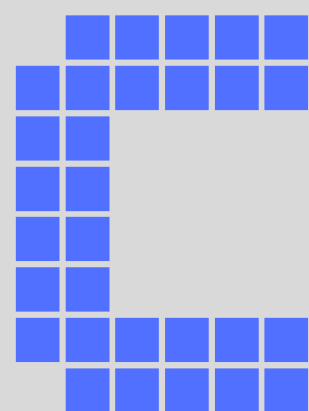
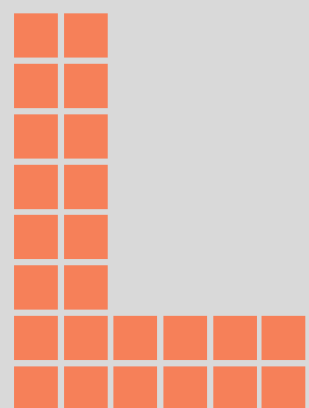
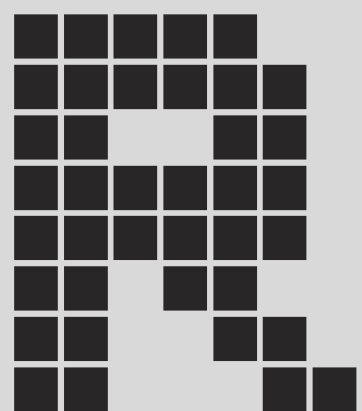
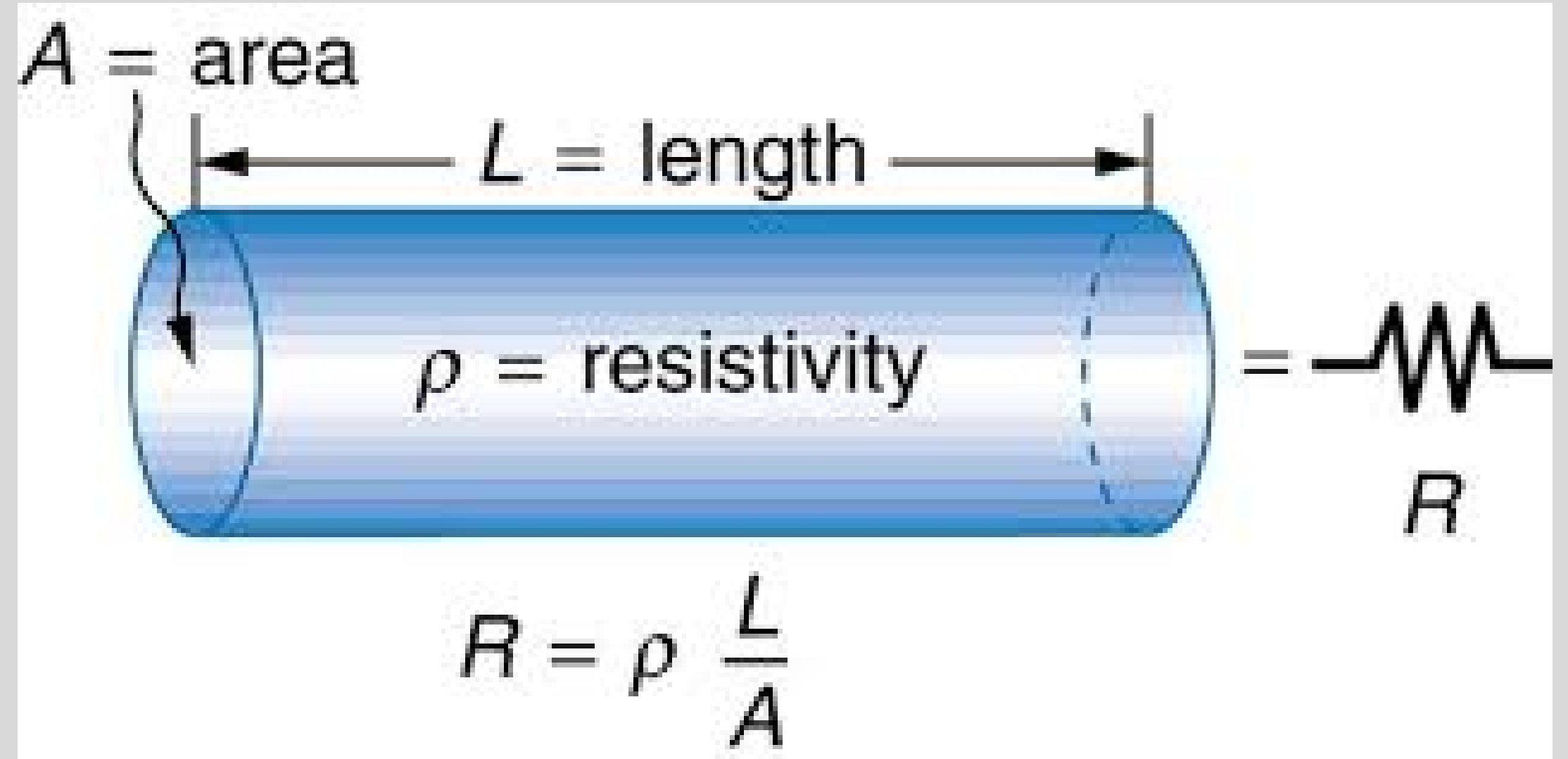
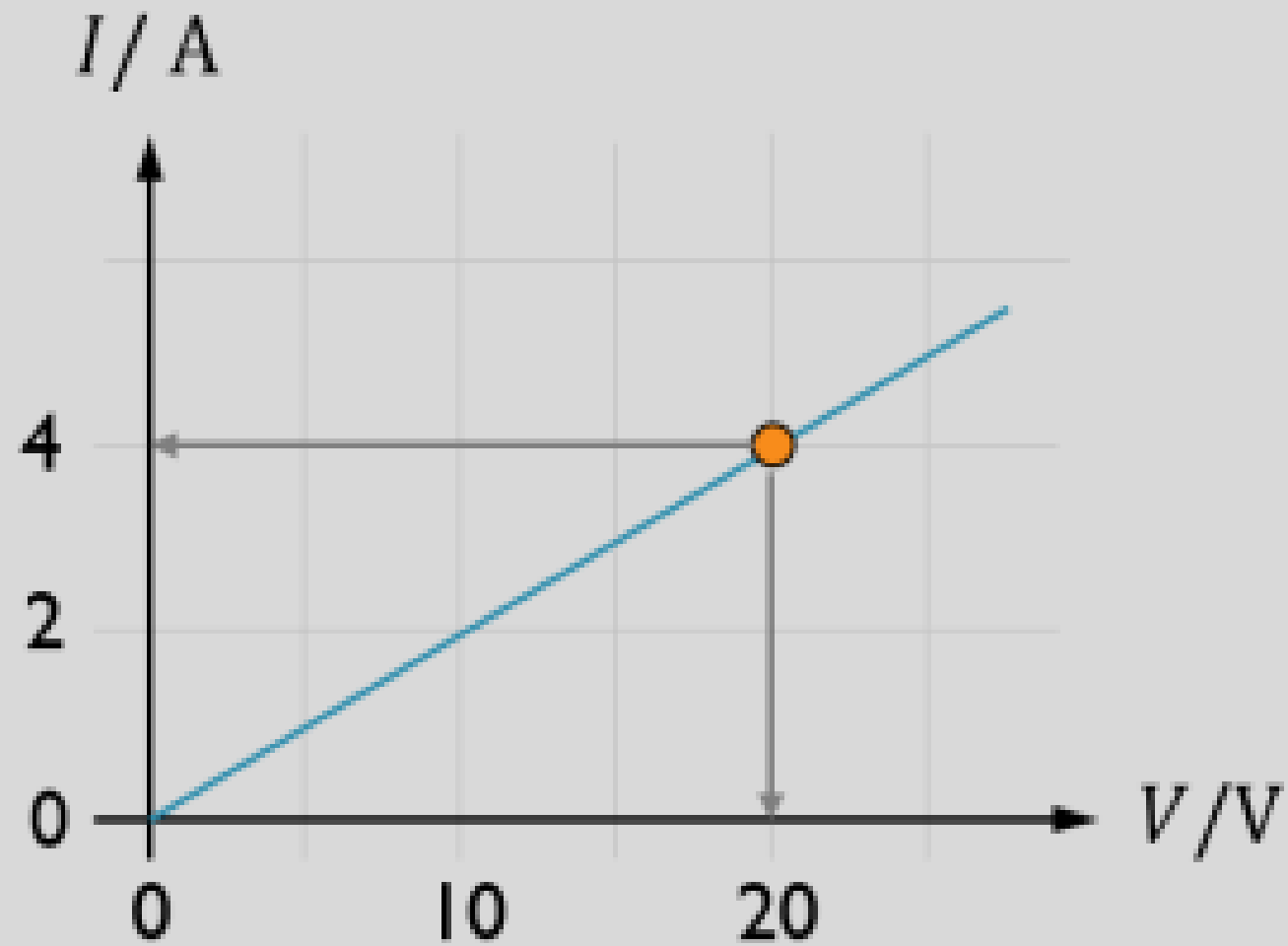
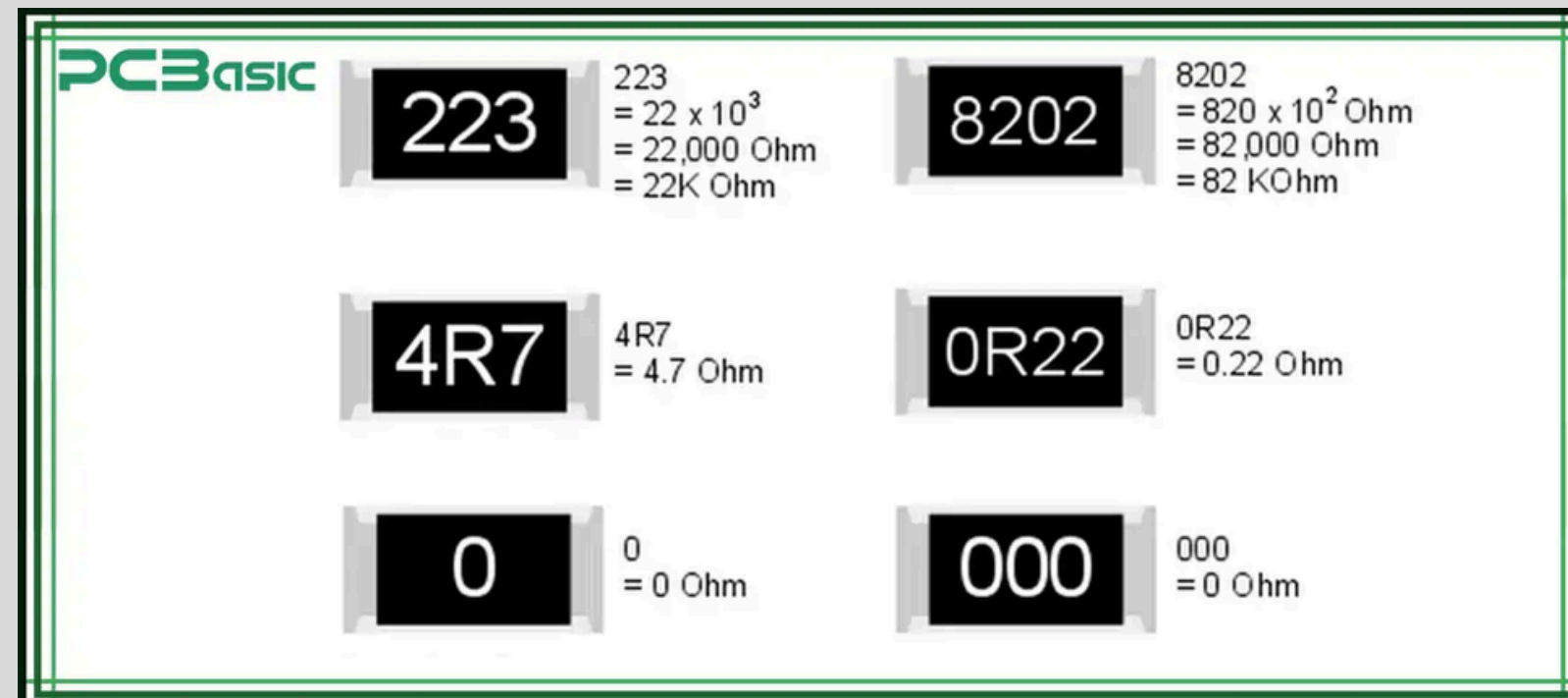
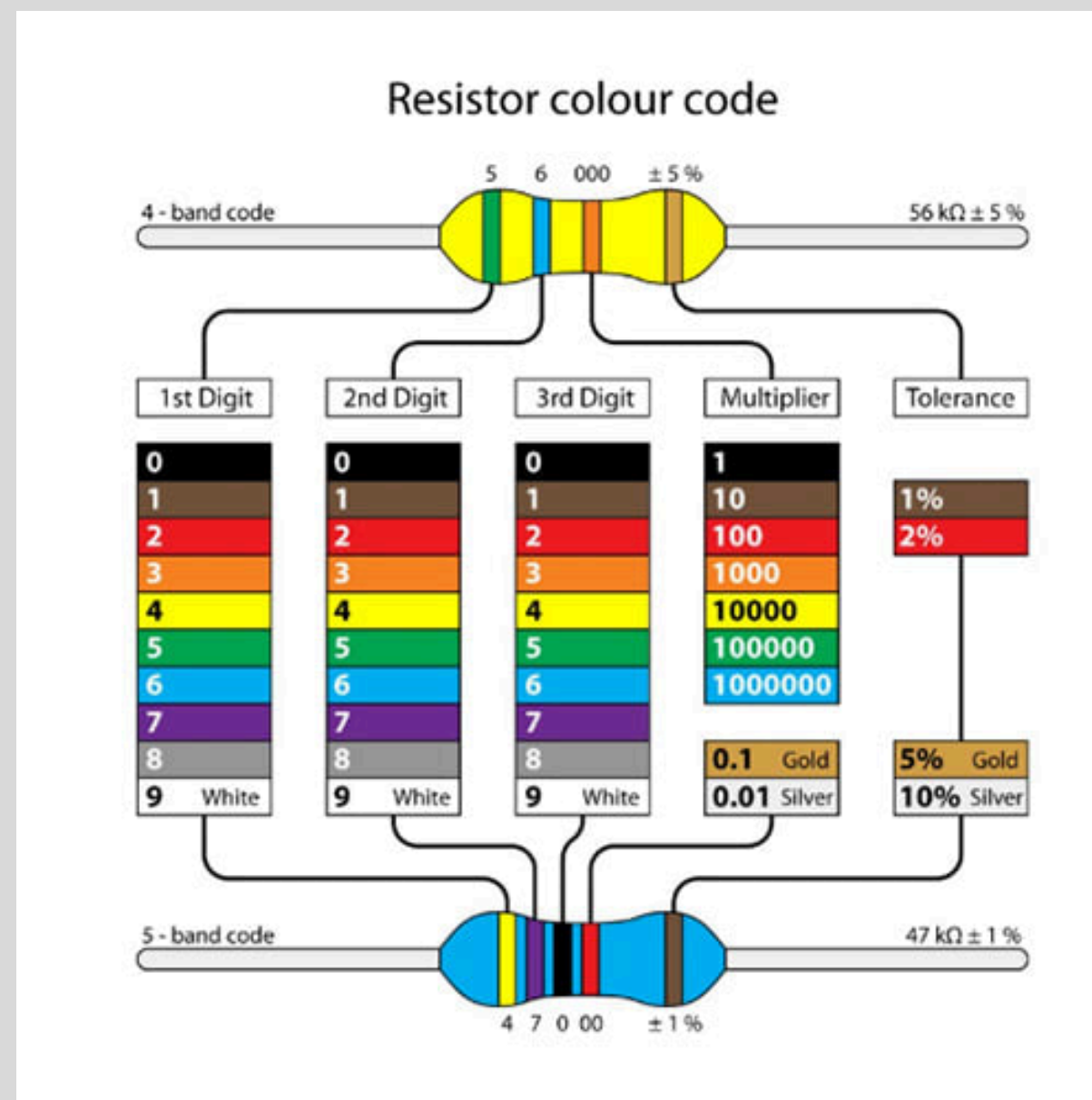
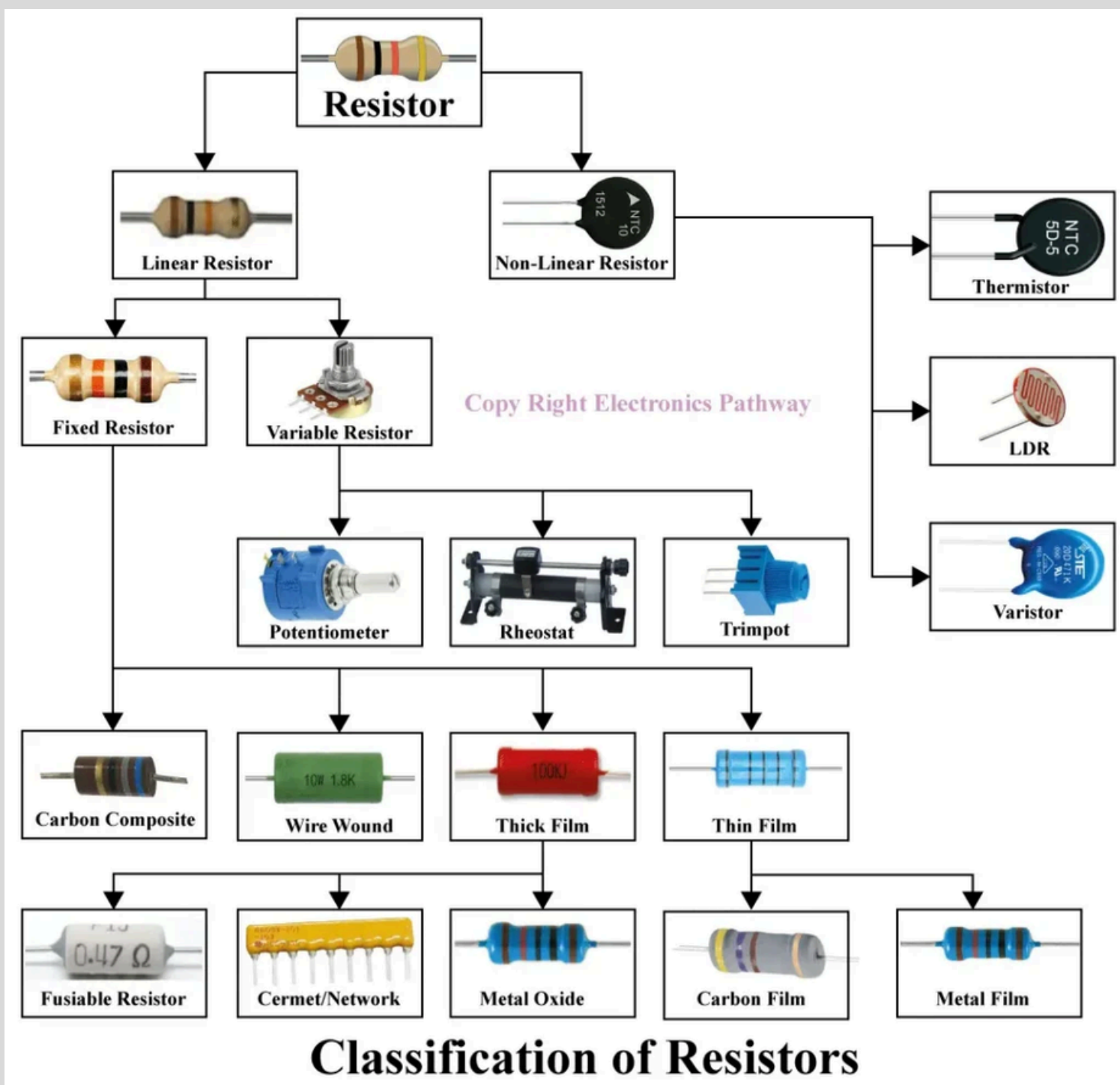


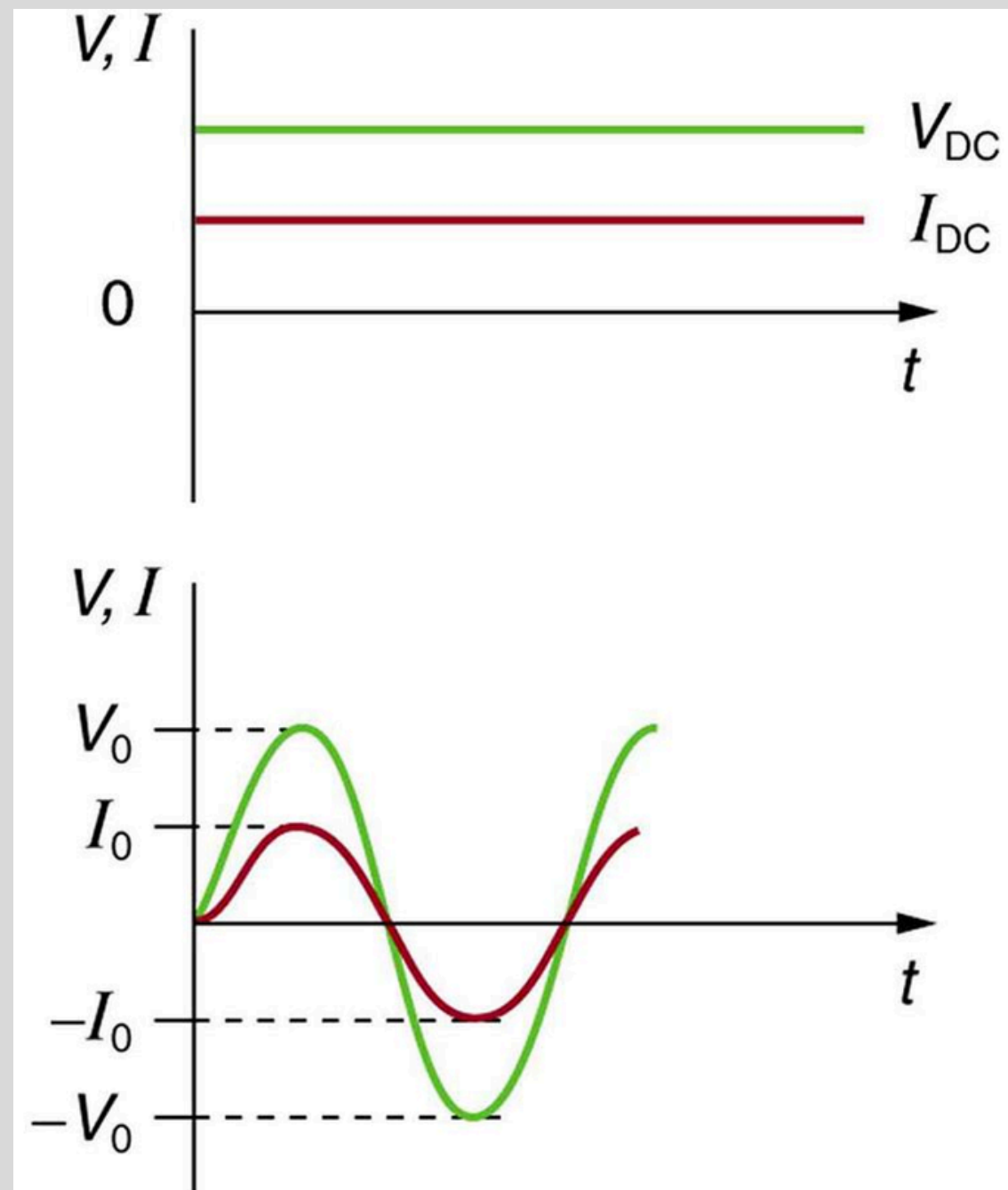


RESISTORS



$$V = IR$$





with DC

with AC

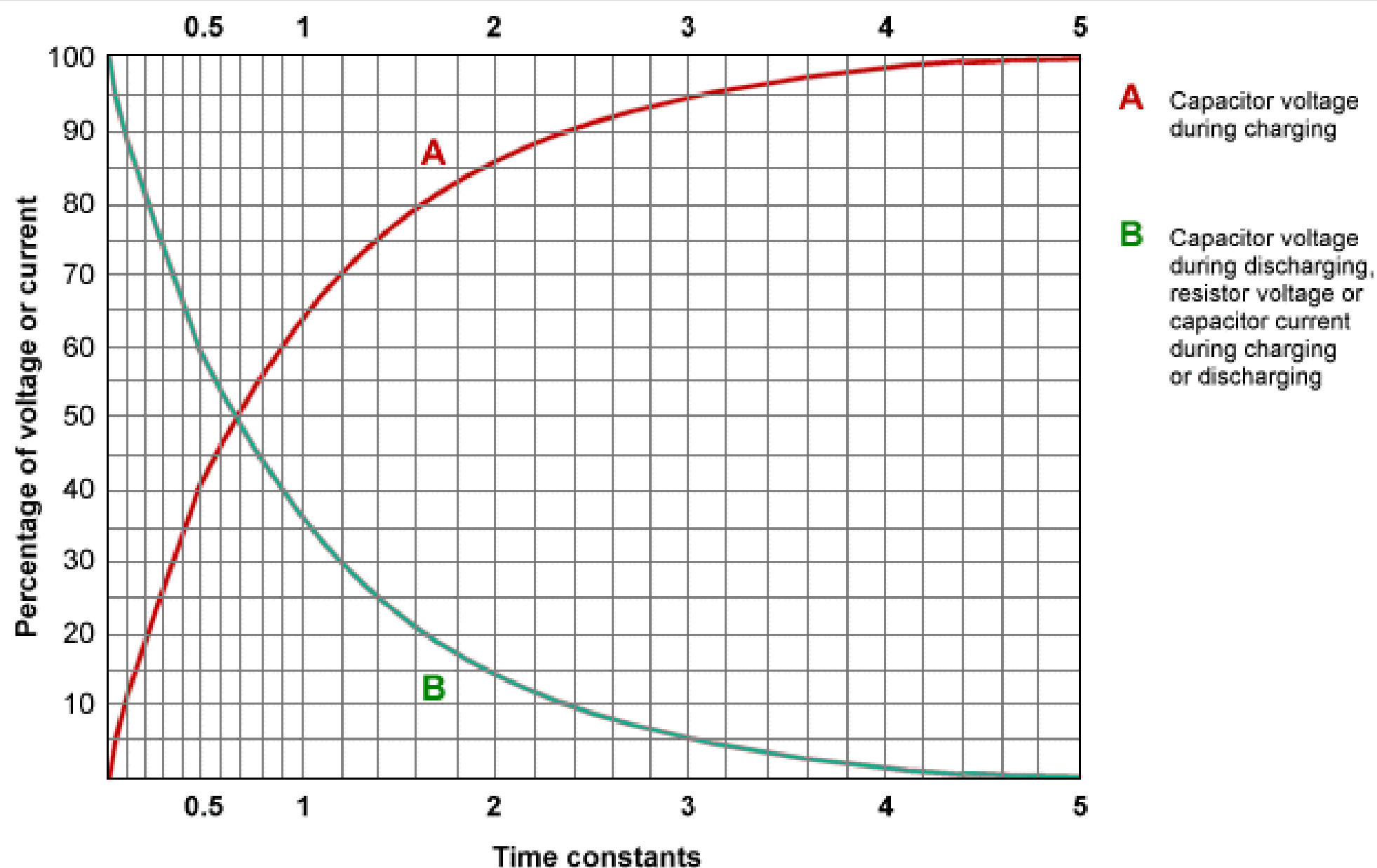
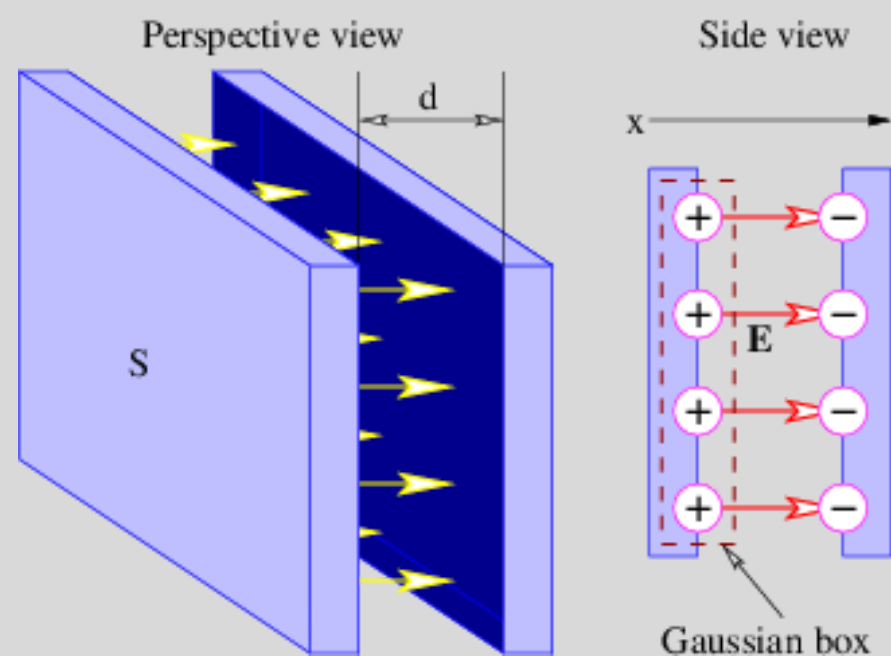
CAPACITORS



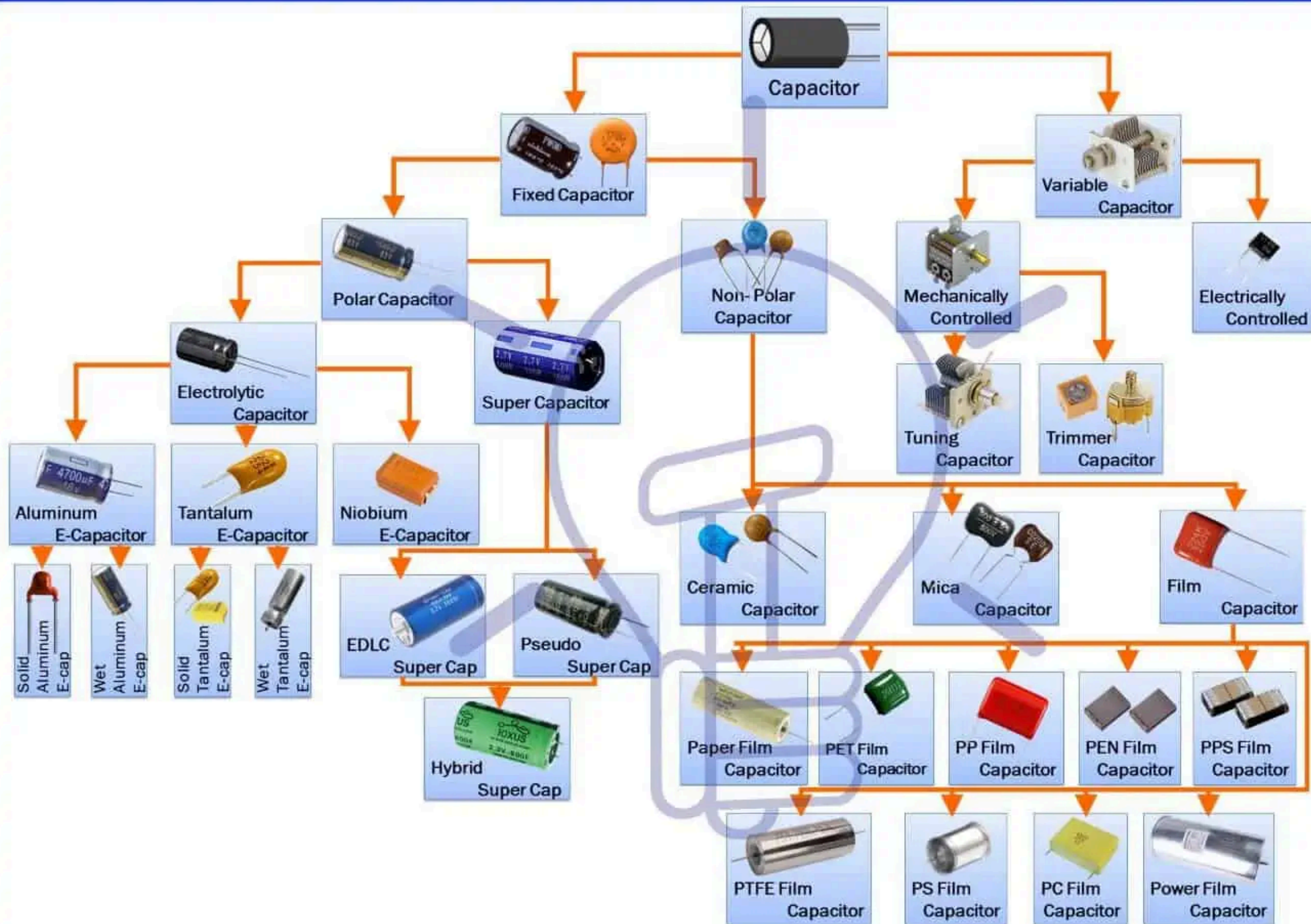
$$\frac{V}{d} = \frac{\left(\frac{Q}{\epsilon_0}\right)}{S} \quad \dots \quad (02)$$

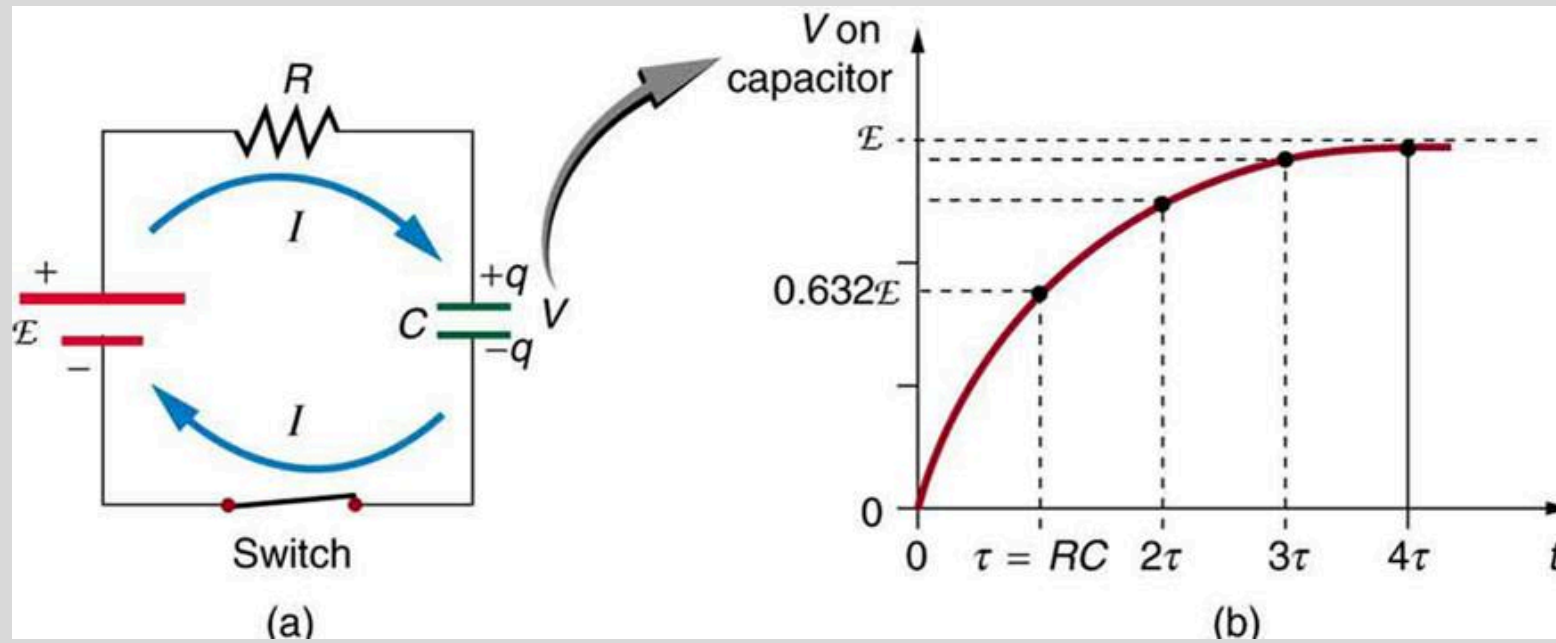
$$Q = \frac{\epsilon_0 \times S \times V}{d} \quad \dots \quad (03)$$

V : applied voltage
 C : capacitance
 Q : electric charge
 ϵ_0 : permittivity of vacuum
 S : area of the flat plate
 d : distance between the plates

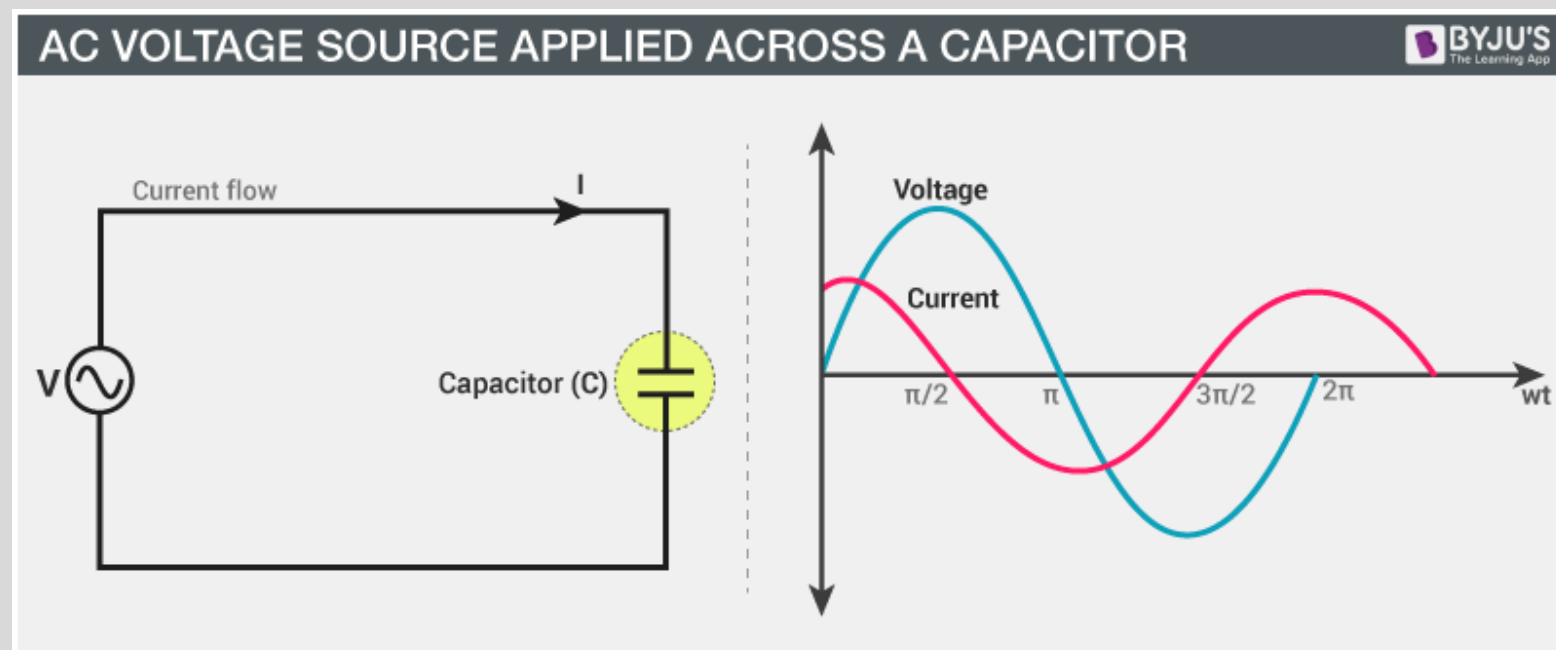


Types Of Capacitors





In a DC circuit, after the capacitor is fully charged, it acts as an open circuit, and current doesn't flow in the circuit



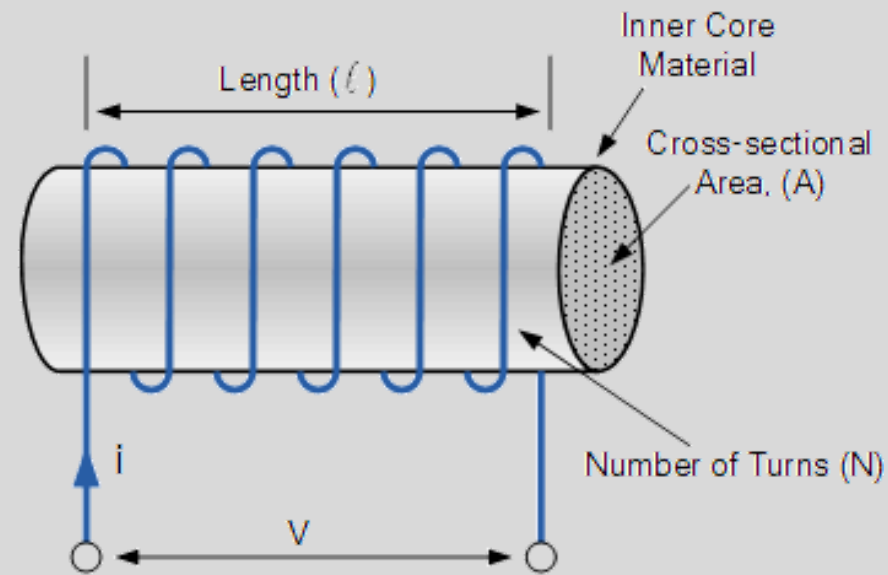
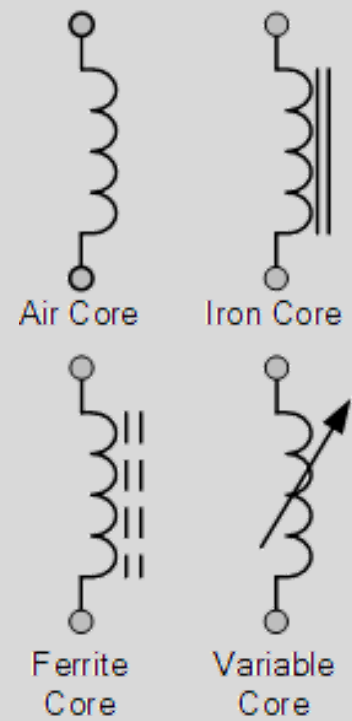
In AC circuits, a capacitor allows current to flow by continuously charging and discharging, causing current to lead voltage by 90° .

$$X_C = \frac{1}{2\pi f C}$$

INDUCTORS

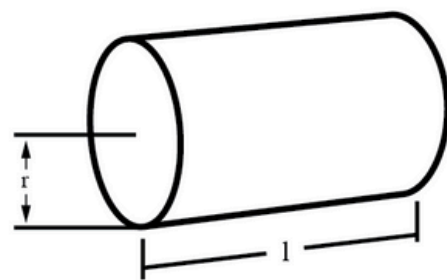


Inductor Symbols



$$L = \frac{N^2 \mu A}{l}$$

$$\mu = \mu_r \mu_0$$



Where,

L = Inductance of coil in Henrys

N = Number of turns in wire coil (straight wire = 1)

μ = Permeability of core material (absolute, not relative)

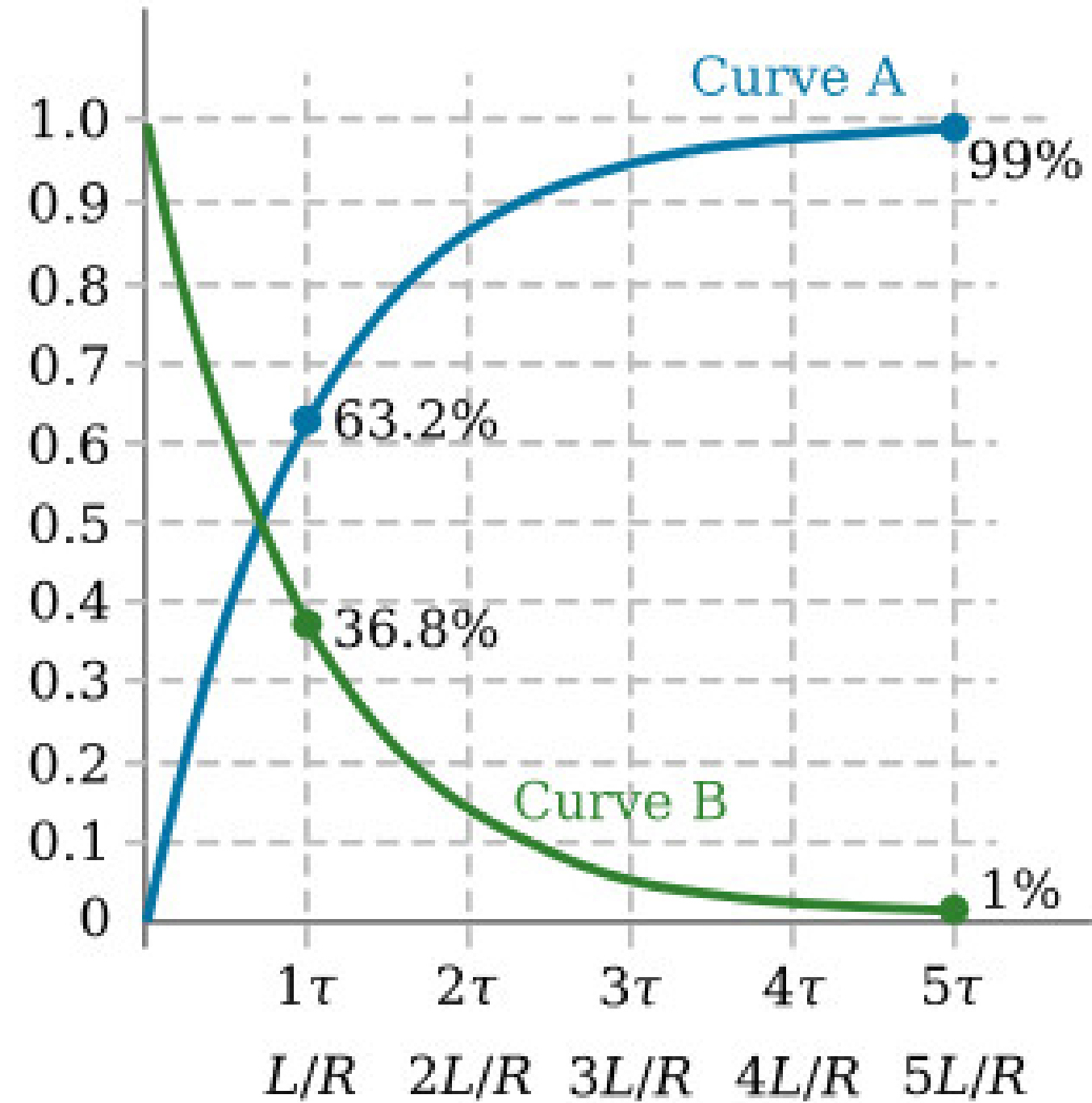
μ_r = Relative permeability, dimensionless ($\mu_0 = 1$ for air)

$\mu_0 = 1.26 \times 10^{-6}$ T-m/At permeability of free space

A = Area of coil in square meters = πr^2

l = Average length of coil in meters

Fraction of maximum i or v ($1 = 100\%$)





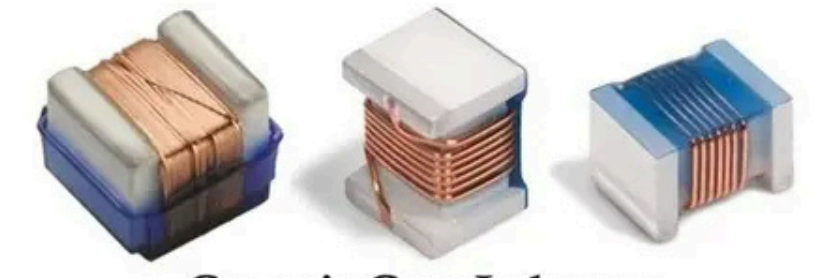
Air Core Inductors



Iron Core Inductors



Ferrite Core Inductors



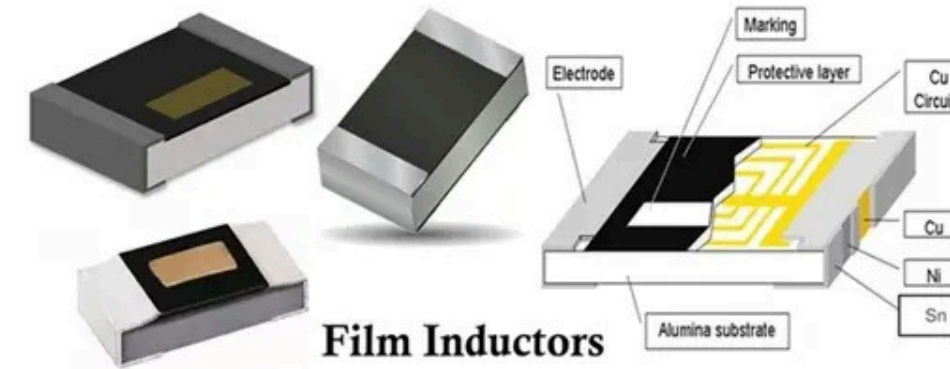
Ceramic Core Inductors



Powdered Iron Core Inductors



Laminated Core Inductors



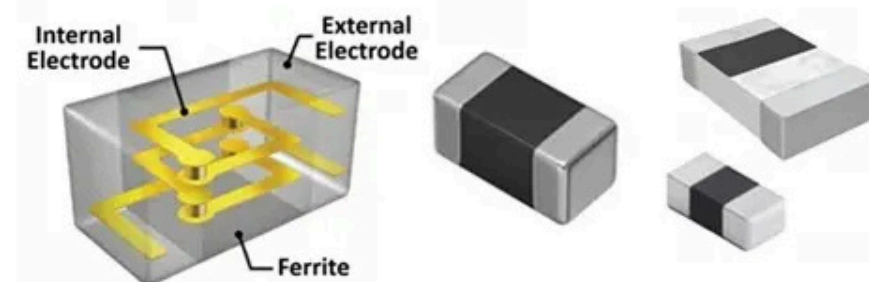
Film Inductors



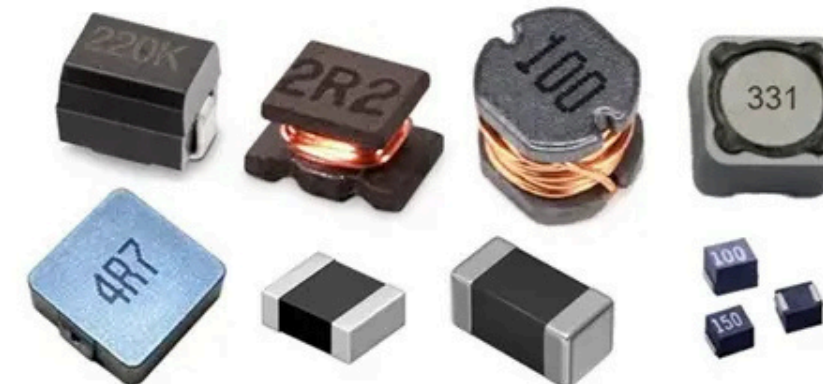
Axial Core Inductors



Bobbin Inductors



Multilayer Chip Inductors



Surface Mount (SMD) Inductors



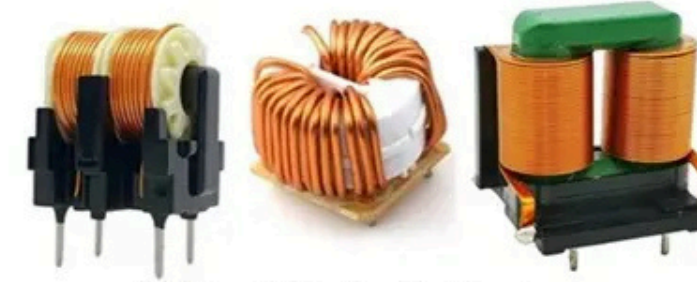
Toroidal Inductors



RF Inductors



Power Inductors



Filter Choke Inductors



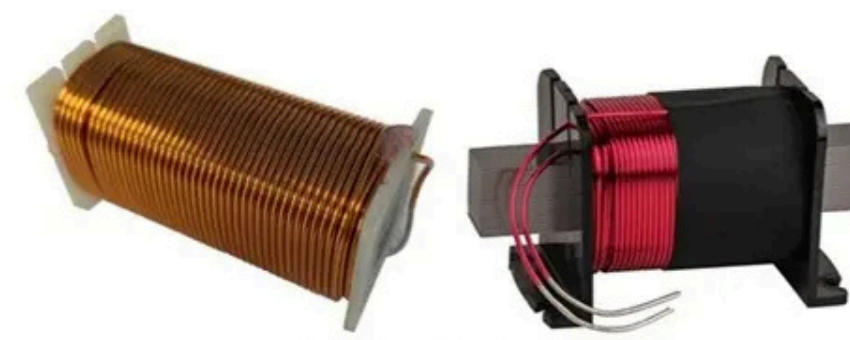
Variable Inductors



Molded Inductors



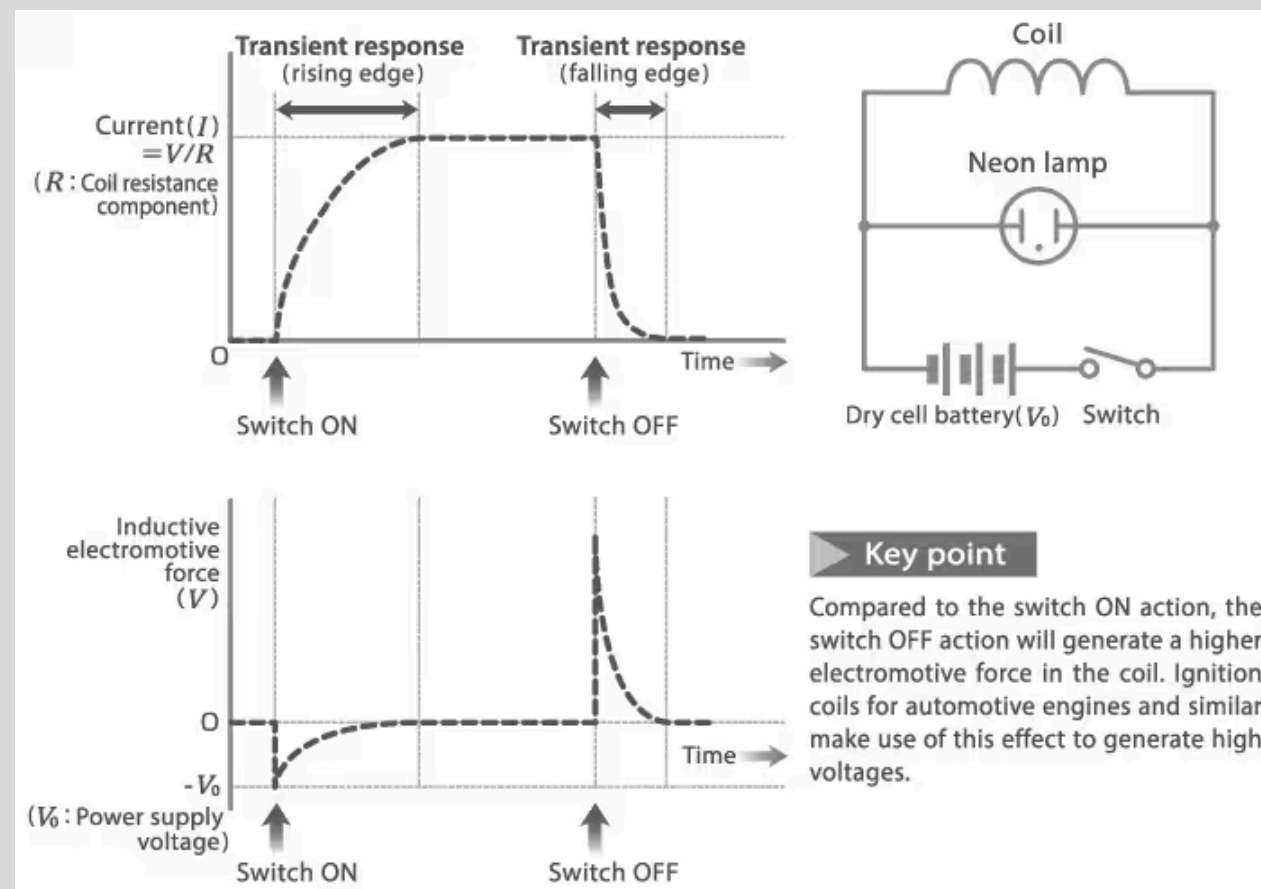
Current Sensing Inductors



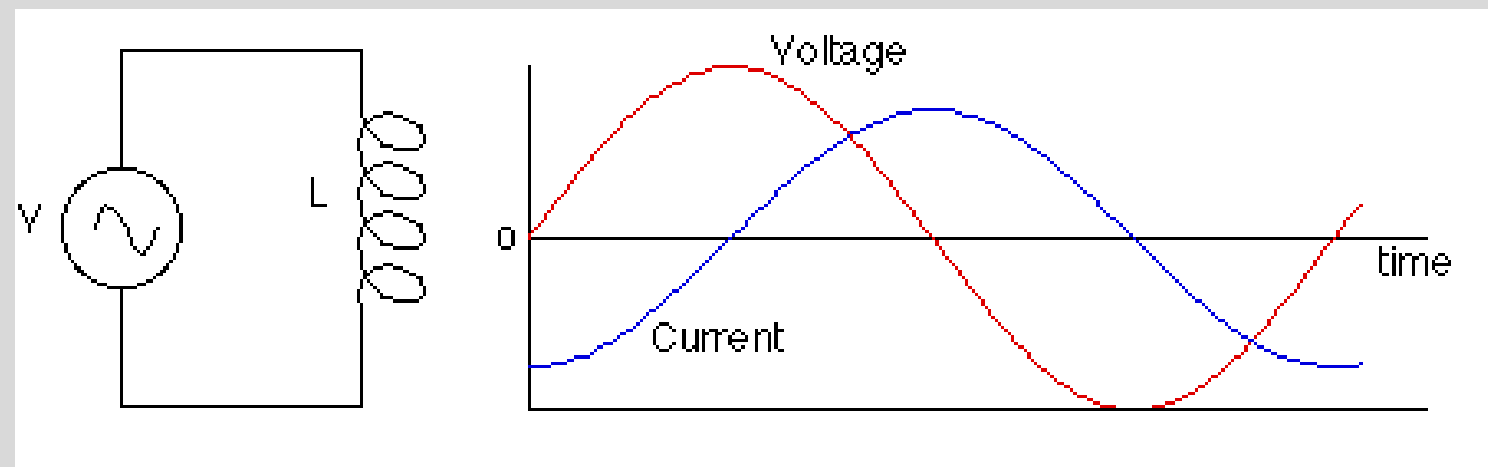
Audio Inductors



Coupling Inductors



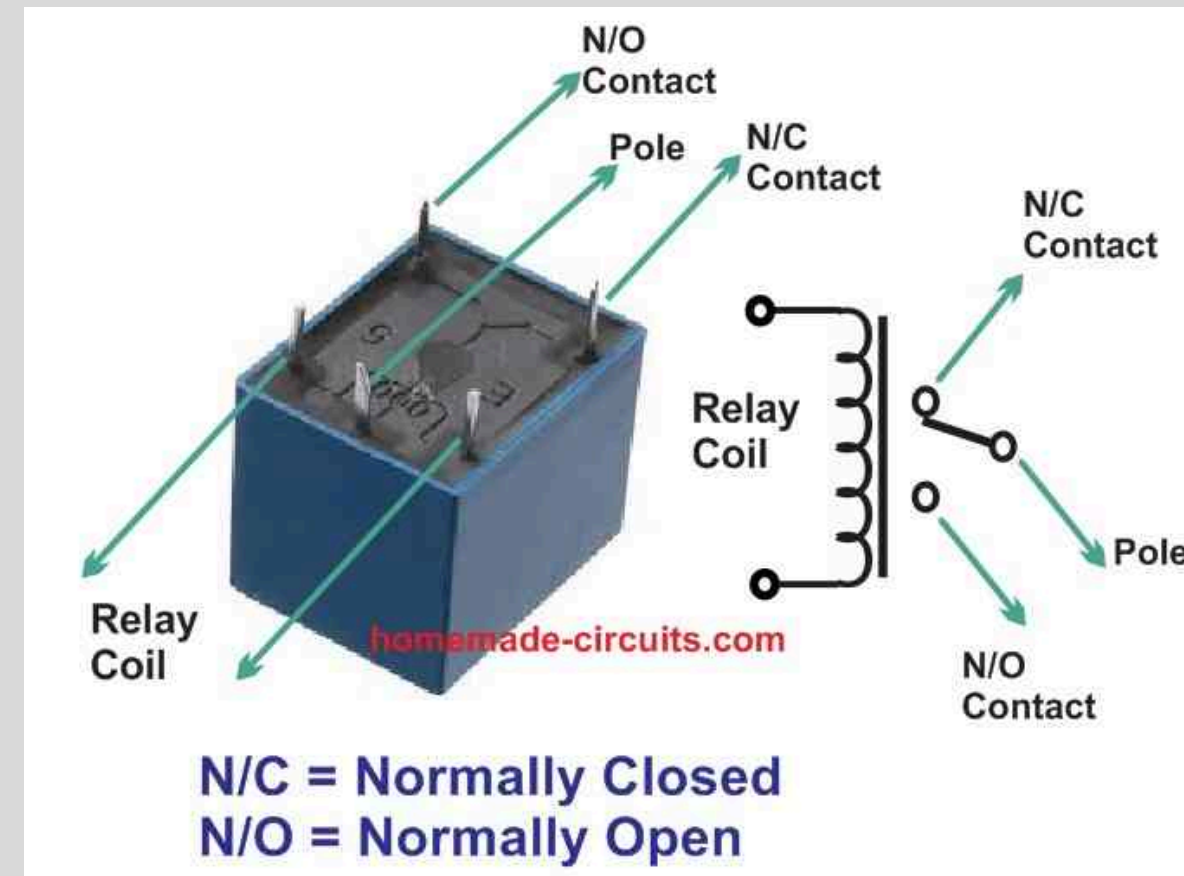
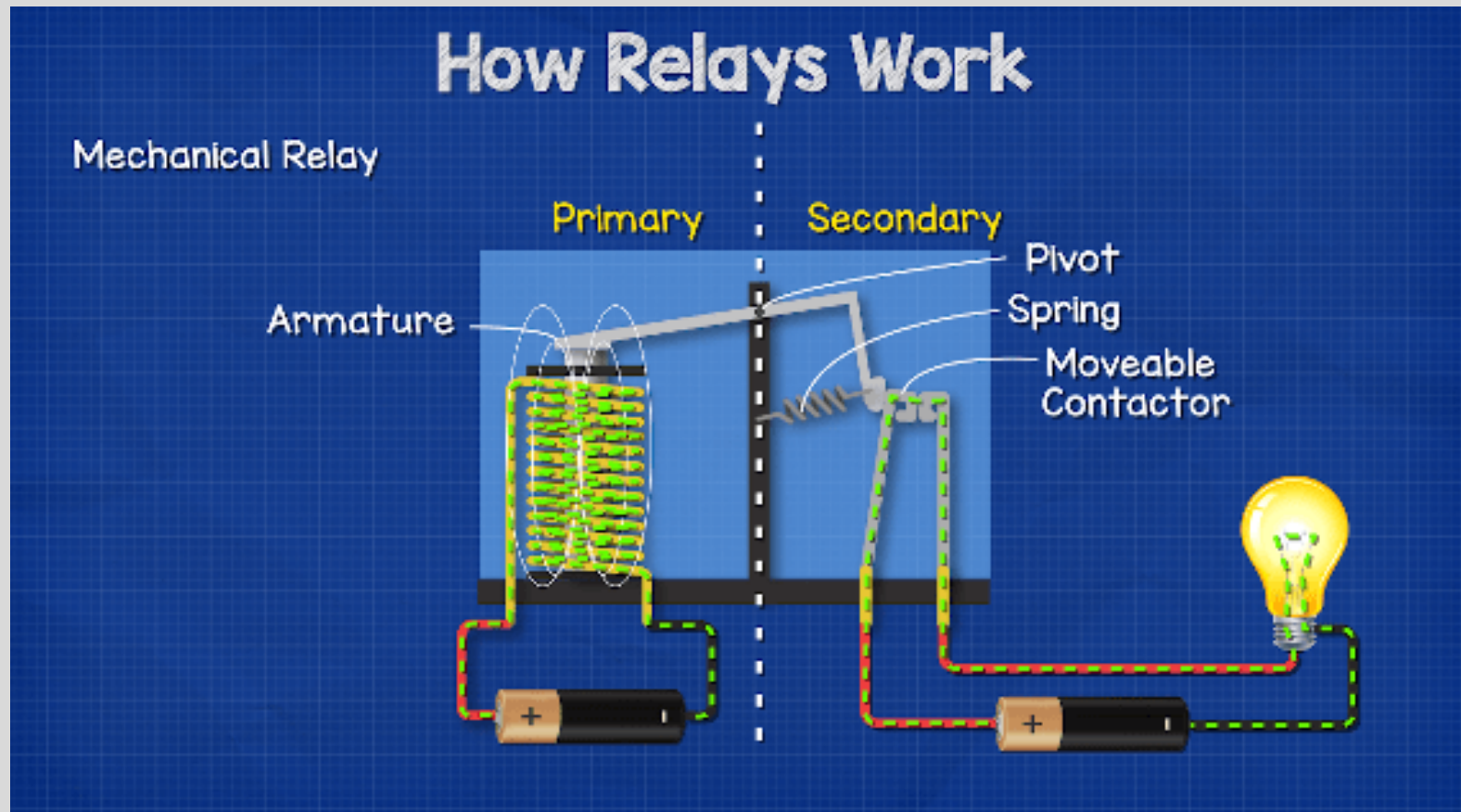
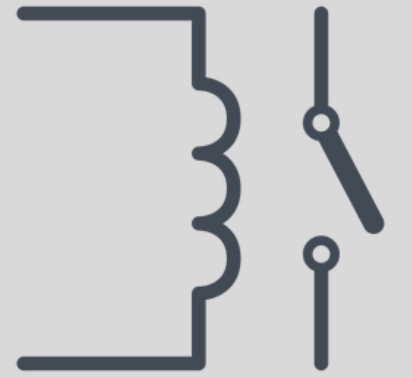
In a DC circuit, a coil prevents an instant change in current, so when the circuit is closed, the coil causes the current change slowly. When the current reaches its max, the coil acts as close circuit



In AC circuits, a coil resists changes in current, causing current to lag behind voltage by 90° .

$$X_L = 2\pi fL$$

RELAY



A relay is an electrically operated switch controlled by a small current. When the relay coil is energized, it produces a magnetic field that pulls the armature. This movement changes the contact position, closing the normally open (NO) contact and opening the normally closed (NC) contact. In this way, a low-power control circuit can safely switch a high-power load.

DIFFERENT TYPES OF RELAY



LATCHING RELAYS



REED RELAY



POLARISED RELAY



BUCHHOLZ RELAY



**OVERLOAD
PROTECTION RELAY**



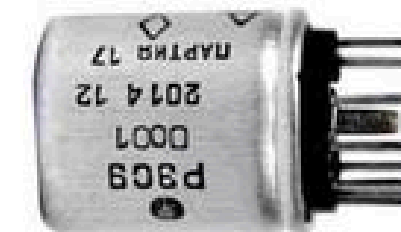
DIFFERENTIAL RELAY



SOLID STATE RELAY



IDMT RELAY



**NON-POLARISED
RELAY**

THANK YOU